

Problem A. Sixteen

Input file: **standard input**
 Output file: **standard output**

You are given an integer x . Your task is to construct a square integer matrix whose determinant is exactly x .

More formally, for each given integer x , you have to output an integer n and an $n \times n$ matrix A such that:

- $1 \leq n \leq 16$;
- every element of A is an integer between -16 and 16 , inclusive;
- $\det(A) = x$.

If there are multiple valid answers, you may output any of them.

Input

The first line contains one integer T ($1 \leq T \leq 100$) — the number of test cases.

Each of the next T lines contains one integer x ($-2^{75} < x < 2^{75}$).

Output

For each test case, output a valid construction in the following format.

First, output one integer n ($1 \leq n \leq 16$) — the size of the matrix.

Then output n lines. The i -th of these lines must contain n integers $a_{i,1}, a_{i,2}, \dots, a_{i,n}$ ($-16 \leq a_{i,j} \leq 16$) — the elements of the matrix.

The determinant of the printed matrix must be equal to the corresponding integer x .

If there are multiple valid answers, you may output any of them.

Example

standard input	standard output
5	1
1	1
2	2
22	0 1
-32	-2 3
138	2
	7 5
	-3 1
	3
	-1 0 2
	3 4 -2
	1 1 7
	3
	-2 12 -1
	-1 3 4
	1 7 4

Problem B. Quadratic Residue

Input file: `standard input`
Output file: `standard output`

Given a positive integer p , your task is to find three positive integers x_1, x_2 , and q such that:

- $1 \leq x_1 < q, 1 \leq x_2 < p$;
- $x_1^2 \equiv p \pmod{q}$;
- $x_2^2 \equiv q \pmod{p}$.

Here, $a \equiv b \pmod{c}$ means the remainder of a divided by c is the same as the remainder of b divided by c .

If multiple solutions exist, any valid set of integers (x_1, x_2, q) is acceptable. If no such integers exist, report that it is impossible.

Input

The first line contains an integer T ($1 \leq T \leq 10^4$), the number of test cases.

Each of the following T lines contains a single integer p ($2 \leq p \leq 10^9$).

Output

For each test case, output a single line:

If a solution exists, output three integers x_1, x_2 , and q ($1 \leq q \leq 10^{12}$, $1 \leq x_1 < q, 1 \leq x_2 < p$) separated by spaces.

If no solution exists, output the word **Impossible**.

Example

standard input	standard output
3	12 1 71
2	19 2 89
5	12 3 69
6	

Problem C. Retest Queue

Input file: `standard input`
Output file: `standard output`

After a large contest, the judging cluster has to perform one more review: it will rerun n programs on the same set of m tests. Every program uses the same queue of tests.

When the judging machine handles one program, it runs the tests from the front of the queue. As soon as some test gives a rejected result, the machine stops processing this program immediately after finishing that test. A program accepted by all tests runs through the whole queue.

Before the review starts, historical logs already contain the behavior of every program on every test. Program i takes $d_{i,j}$ time to run on test j , and the result is either accepted or rejected. Whenever this test is actually run, $d_{i,j}$ is added to the total time of the cluster, regardless of the result.

You may reorder the m tests arbitrarily before the review. Find the minimum total time needed to finish judging all n programs.

Input

The first line contains two integers n and m , the number of programs and the number of tests.

The following n lines describe the programs. The i -th of these lines contains m integers $d_{i,1}, d_{i,2}, \dots, d_{i,m}$, followed by a string s_i of length m . If the j -th character of s_i is A, program i is accepted on test j ; if it is R, program i is rejected on test j .

It is guaranteed that $1 \leq n \leq 2 \times 10^4$, $1 \leq m \leq 20$, $1 \leq d_{i,j} \leq 10^9$, and each s_i consists only of characters A and R.

Output

Output one integer: the minimum possible total time.

Example

standard input	standard output
3 3 3 10 2 ARA 5 1 4 RAA 2 2 7 AAR	27

Note

One optimal order in the sample is to run test 2 first, then test 1, then test 3.

With this order, the three programs take 10, $1 + 5 = 6$, and $2 + 2 + 7 = 11$ time respectively, for a total of 27.

Problem D. The Game

Input file: **standard input**
 Output file: **standard output**

Alice and Bob are playing a game.

Initially, there is an empty sequence. They are given an integer n , and they will construct a permutation p of length n .

Alice moves first. On each move, the current player chooses one integer that has not been chosen before from 1 to n , and appends it to the end of the sequence. After exactly n moves, the sequence becomes a permutation p of $1, 2, \dots, n$.

A cyclic shift of a permutation $p = (p_1, p_2, \dots, p_n)$ is a sequence obtained by choosing an index i ($1 \leq i \leq n$) and writing $(p_i, p_{i+1}, \dots, p_n, p_1, p_2, \dots, p_{i-1})$. For example, the cyclic shifts of $(2, 3, 1)$ are $(2, 3, 1)$, $(3, 1, 2)$, and $(1, 2, 3)$.

For a permutation p , define $f(p)$ as the lexicographically smallest cyclic shift of p .

Alice wants to minimize $f(p)$ lexicographically, while Bob wants to maximize it lexicographically.

Assuming both players play optimally, determine the resulting permutation $f(p)$.

Input

The first line contains an integer T ($1 \leq T \leq 10^5$), the number of test cases.

Each of the next T lines contains one integer n ($1 \leq n \leq 5 \cdot 10^5$).

It is guaranteed that the sum of n over all test cases does not exceed $5 \cdot 10^5$.

Output

For each test case, output n integers: the resulting permutation $f(p)$ when both players play optimally.

Example

standard input	standard output
4	1
1	1 2
2	1 3 2
3	1 3 2 4
4	

Problem E. DPRS

Input file: **standard input**
 Output file: **standard output**

You are given a strongly connected weighted directed graph with n vertices and m directed edges. The vertices are numbered from 1 to n , and the edges are numbered from 1 to m .

The k -th edge is denoted by (u_k, v_k, w_k) , which means there is a directed edge from u_k to v_k with weight w_k .

Let $dis(i, j)$ denote the length of the shortest path from vertex i to vertex j .

For a graph, define its value as

$$\max_{\substack{1 \leq i, j \leq n \\ i \neq j}} \frac{dis(i, j)}{dis(j, i)}.$$

There are q queries. In each query, you are given an edge index k and a new weight x , where $1 \leq x < w_k$. For this query only, the weight of the k -th edge is temporarily changed to x . You need to output the value of the graph after this temporary modification.

All queries are independent. That is, after each query, the graph is restored to its original state.

Input

The first line contains an integer T ($1 \leq T \leq 1000$), the number of test cases.

For each test case, the first line contains three integers n, m, q ($2 \leq n \leq m \leq 2000$, $1 \leq q \leq 2000$).

The sum of n , the sum of m , and the sum of q over all test cases do not exceed 2000. That is, $\sum n \leq 2000$, $\sum m \leq 2000$, and $\sum q \leq 2000$.

The next m lines each contain three integers u_k, v_k, w_k , describing the k -th directed edge, where $1 \leq u_k, v_k \leq n$, $u_k \neq v_k$, and $1 \leq w_k \leq 10^9$.

The graph in each test case is guaranteed to be strongly connected.

The next q lines each contain two integers k, x , describing one query, where $1 \leq k \leq m$ and $1 \leq x < w_k$.

In this query, the weight of the k -th edge is temporarily changed to x .

Output

For each query, output one line containing one real number, representing the maximum value of $\frac{dis(i, j)}{dis(j, i)}$ over all ordered pairs (i, j) with $i \neq j$.

Your answer will be considered correct if its absolute or relative error does not exceed 10^{-6} .

Example

standard input	standard output
2	6.000000000000
3 3 2	7.000000000000
1 2 2	6.000000000000
2 3 3	3.750000000000
3 1 4	3.600000000000
2 1	7.500000000000
1 1	
5 7 4	
1 2 7	
1 4 3	
2 3 4	
4 1 2	
1 5 5	
5 2 6	
3 1 8	
5 3	
6 5	
1 4	
3 2	

Problem F. 23 Subsequences

Input file: **standard input**
 Output file: **standard output**

Define a sequence $b = (b_1, b_2, \dots, b_m)$ to be good if and only if for every $i = 2, 3, \dots, m$,

$$2 \cdot b_{i-1} \leq b_i \leq 3 \cdot b_{i-1}.$$

In particular, any sequence of length 1 is always good.

Now, given a sequence of positive integers $a = (a_1, a_2, \dots, a_n)$ of length n , you need to answer q queries. Each query gives an interval $[l, r]$ ($1 \leq l \leq r \leq n$). For each query, find the length of the longest good subsequence in the subsequence $a_l, a_{l+1}, \dots, a_r$, while preserving the original order.

Note: The subsequence does not have to be contiguous, but it must preserve the relative order of the original sequence.

Input

The first line contains two integers n, q ($1 \leq n, q \leq 2 \cdot 10^5$), denoting the length of the sequence and the number of queries, respectively.

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^{18}$).

The next q lines each contain two integers l, r ($1 \leq l \leq r \leq n$), denoting the interval of a query.

Output

Output q lines, each containing one integer, the answer to each query in order.

Example

standard input	standard output
5 4	3
1 2 4 6 17	2
1 3	4
2 4	3
1 5	
1 4	

Problem G. Fading Memories

Input file: `standard input`
 Output file: `standard output`

Please remember that I was once here—and
 then, slowly, let the memory fade.
 But look: those words that lingered
 unspoken, those gestures still suspended in
 midair, have all remained here, as though
 they were never truly brought to an end.
 The wind will still pass through this street,
 stirring the hem of your clothes, and carrying
 back some distant fragment of the past—
 If everything is destined to fade away, could
 we walk this road once more?

— Anonymous

Given a sequence b of length n .

Define an interval $[l, r]$ to be good if and only if for all $l \leq i \leq r$,

$$\min(b_l, b_r) \leq b_i \leq \max(b_l, b_r).$$

You may perform the following operation any number of times:

Choose a good interval $[l, r]$, and delete all elements whose indices are between $l + 1$ and $r - 1$. After the deletion, the length of the sequence decreases, and the original elements at positions l and r become adjacent. All subsequent operations are performed on the new sequence.

Your goal is to make the final length of the sequence as small as possible by choosing an optimal order of operations. This minimum possible length is called the eliminated length of the sequence.

Now you need to process q events, each of one of the following two types:

- 1 x y — change b_x to y ;
- 2 L R — take the subarray $b[L \dots R]$ as the initial sequence, and ask for its eliminated length.

Please output the answers to all queries.

Input

The first line contains two integers n, q ($1 \leq n, q \leq 2 \cdot 10^5$).

The second line contains n integers $b_1, b_2, \dots, b_n$ ($1 \leq b_i \leq 10^9$).

The next q lines each begin with an integer opt , indicating the type of operation:

- If $opt = 1$, it is followed by two integers x, y ($1 \leq x \leq n, 1 \leq y \leq 10^9$);
- If $opt = 2$, it is followed by two integers L, R ($1 \leq L \leq R \leq n$).

Examples

standard input	standard output
6 5 3 1 4 1 5 9 2 1 6 1 4 2 2 1 6 1 3 8 2 2 5	3 3 4
10 6 9 8 8 2 7 1 8 9 4 3 2 1 10 1 4 9 2 1 10 1 3 7 1 5 8 2 3 10	4 4 5

Problem H. String

Input file: **standard input**
Output file: **standard output**

You are given n strings $s_1, s_2, \dots, s_n$.

You may reorder these n strings in any way and then concatenate them in that order to obtain a string S .

For each integer k from 0 to L (where $L = \sum_{i=1}^n |s_i|$) independently, you can reorder the n strings in any way, concatenate them in that order to obtain a string S , and then change at most k characters in S . In one change, you replace one character of S with another lowercase English letter.

Your task is to find, for each k from 0 to L , the lexicographically smallest string that can be obtained.

Input

The first line contains an integer T ($1 \leq T \leq 500$), denoting the number of test cases.

For each test case, the first line contains one integer n ($1 \leq n \leq 500$).

The second line contains n non-empty strings $s_1, s_2, \dots, s_n$, separated by spaces. Each string consists only of lowercase English letters.

For each test case, let $L = \sum_{i=1}^n |s_i|$.

It is guaranteed that the sum of n over all test cases does not exceed 500, and the sum of L over all test cases does not exceed 500.

Output

For each test case, output $L + 1$ lines.

For the k -th line, where $0 \leq k \leq L$, output the lexicographically smallest string that can be obtained by changing at most k characters.

The answers for each test case must be printed in increasing order of k , from 0 to L .

Example

standard input	standard output
3	aabbcaczz
5	aaabbcazz
bca a zz ab c	aaaaabczz
4	aaaaaabzz
ba b aa aba	aaaaaaabc
3	aaaaaaaab
az za m	aaaaaaaaa
	aaaaaaaaa
	aaaaaaaaa
	aaaaaaaaa
	aaababab
	aaaaabab
	aaaaaaaab
	aaaaaaaaa
	aaaaaaaaa
	aaaaaaaaa
	aaaaaaaaa
	azmza
	aaazm
	aaaam
	aaaaa
	aaaaa
	aaaaa

Problem I. Rounddog II

Input file: **standard input**
Output file: **standard output**

You are a crazy fan of Rounddog, so you're going to come up with a problem about Rounddog.

Let $T_1 = \text{Rounddog}$.

For every integer $i > 1$, define $T_i = T_{i-1} + \mathbf{g}$, where $+$ denotes string concatenation.

In other words, T_i is the string **Rounddo** followed by exactly i characters **g**.

You are given a string S and an integer k . Consider all cyclic shifts of S .

A cyclic shift of S is obtained by choosing an index and moving the prefix before this index to the end of the string. For example, the cyclic shifts of **abcd** are **abcd**, **bcda**, **cdab**, and **dabc**.

Your task is to count how many cyclic shifts of S contain T_k as a contiguous substring.

Input

The first line contains an integer t ($1 \leq t \leq 10^5$), denoting the number of test cases.

For each test case, the first line contains a string S ($1 \leq |S| \leq 10^5$).

The second line contains an integer k ($1 \leq k \leq 100$).

It is guaranteed that the sum of $|S|$ over all test cases does not exceed 10^5 , and S only contains uppercase letters and lowercase letters.

Output

For each test case, output one integer, denoting the number of cyclic shifts of S that contain T_k as a contiguous substring.

Example

standard input	standard output
3	12
Rounddogggggggggggggggggg	8
6	0
RounddogRounddo	
1	
abcdefgabcdegf	
2	

Problem J. Walk

Input file: **standard input**
 Output file: **standard output**

You are given a simple connected undirected graph with n vertices and m edges, and a sequence of special vertices $x_1, x_2, \dots, x_L$.

You want to start from x_1 and visit $x_1, x_2, \dots, x_L$ in this order using a DFS-like process (you can visit extra vertices, but $x_1, x_2, \dots, x_L$ must be visited in order).

The process is defined by a stack. Initially, the stack contains only x_1 , and x_1 is considered visited. In one operation, you may do exactly one of the following:

- Let u be the current top of the stack. Push a vertex v onto the stack, where v is adjacent to u in the graph. The vertex v is considered visited.
- Pop the current top vertex from the stack. If the stack is not empty after the pop, the new top vertex is considered visited.

The final content of the stack does not matter.

Among all valid processes that visit $x_1, x_2, \dots, x_L$ in order using the minimum possible number of operations, find the minimum possible value of the maximum stack size that appears during the process.

If there are consecutive vertices that are same in the array x , we consider them all as visited once you visit the first vertex.

It can be shown that the minimum possible number of operations is $\sum_{i=1}^{L-1} \text{dist}(x_i, x_{i+1})$, where $\text{dist}(a, b)$ denotes the length of a shortest path between vertices a and b .

Input

The first line contains an integer T ($1 \leq T \leq 60$), the number of test cases.

For each test case, the first line contains three integers n ($2 \leq n \leq 200$), m ($n - 1 \leq m \leq \frac{n(n-1)}{2}$), and L ($1 \leq L \leq 60$), denoting the number of vertices, the number of edges, and the length of the special sequence, respectively.

Each of the next m lines contains two integers u ($1 \leq u \leq n$) and v ($1 \leq v \leq n$), indicating an undirected edge between vertices u and v .

The last line contains L integers $x_1, x_2, \dots, x_L$ ($1 \leq x_i \leq n$).

It is guaranteed that each given graph is simple and connected.

It is also guaranteed that $\sum n \leq 200$ and $\sum L \leq 60$ over all test cases.

Output

For each test case, output a single integer: the minimum possible maximum stack size among all valid processes that use the minimum possible number of operations.

Example

standard input	standard output
4	5
5 4 5	4
1 2	3
2 3	4
3 4	
4 5	
1 3 5 4 2	
6 7 6	
1 2	
2 3	
3 4	
4 5	
5 6	
6 1	
2 5	
1 4 6 3 5 2	
7 6 6	
1 2	
1 3	
1 4	
1 5	
1 6	
1 7	
2 3 4 5 6 7	
5 10 4	
1 2	
1 3	
1 4	
1 5	
2 3	
2 4	
2 5	
3 4	
3 5	
4 5	
1 5 3 2	

Problem K. Decomposition Trees

Input file: **standard input**
 Output file: **standard output**

You are given a tree T with n vertices, numbered from 1 to n .

We define a decomposition tree of T recursively as follows.

For any non-empty connected induced subgraph S of T , choose an arbitrary vertex $c \in S$ as the current decomposition center. The vertex c becomes the root of the decomposition tree of S . Then remove c from S . Each remaining connected component is decomposed independently by the same rule, and the roots of their decomposition trees become children of c .

Unlike the usual centroid decomposition, the chosen center c does not have to be a centroid. Any vertex in the current connected component may be chosen.

Two decomposition trees are considered different if, as rooted trees on the same vertex set $\{1, 2, \dots, n\}$, there exists at least one vertex whose parent is different. In particular, decomposition trees with different roots are always considered different.

Your task is to count the number of different decomposition trees that can be obtained from T .

Since the answer may be very large, output it modulo $10^9 + 7$.

Input

The first line contains one integer n ($1 \leq n \leq 3000$).

Each of the next $n - 1$ lines contains two integers u and v ($1 \leq u, v \leq n$), denoting an edge between vertices u and v .

It is guaranteed that the given graph is a tree.

Output

Output one integer: the number of different decomposition trees of T , modulo $10^9 + 7$.

Examples

standard input	standard output
3 1 2 2 3	5
5 1 2 1 3 2 4 2 5	51
7 1 2 1 4 2 3 2 5 4 6 5 7	605

Problem L. Geometry

Input file: **standard input**
 Output file: **standard output**

You are given a convex polygon A in the plane. The origin $(0, 0)$ is guaranteed to lie strictly inside A .

You are also given m scored points. The i -th point is $p_i = (x_i, y_i)$ and has a positive score a_i .

For a real number $\lambda \geq 0$, define $\lambda A = \{\lambda p : p \in A\}$, that is, every coordinate of every point in A is multiplied by λ with respect to the origin.

You need to find the minimum real number $\lambda \geq 0$ such that it is possible to draw a closed disk C satisfying both of the following conditions:

- $C \subseteq \lambda A$;
- the total score of the given points contained in C is at least W .

A closed disk is a set of the form $C = \{p \in \mathbb{R}^2 : \|p - o\|_2 \leq r\}$, where o is the center of the disk and $r \geq 0$ is its radius.

A point lying on the boundary of the disk is considered to be contained in the disk. The disk is also allowed to touch the boundary of λA .

It is guaranteed that $1 \leq W \leq \sum_{i=1}^m a_i$, so the answer always exists and is finite.

Input

The first line contains an integer T ($1 \leq T \leq 100$), the number of test cases.

For each test case, the first line contains three integers n ($3 \leq n \leq 1000$), m ($1 \leq m \leq 1000$), and W ($1 \leq W \leq \sum_{i=1}^m a_i$).

The next n lines describe the convex polygon A . The j -th of these lines contains two integers X_j and Y_j ($|X_j|, |Y_j| \leq 10^6$), the coordinates of the j -th vertex of A .

The vertices of A are given in counterclockwise order. The polygon is convex, and the origin lies strictly inside it.

The next m lines describe the scored points. The i -th of these lines contains three integers x_i , y_i , and a_i ($|x_i|, |y_i| \leq 10^6$, $1 \leq a_i \leq 10^9$), meaning that the point (x_i, y_i) has score a_i .

It is guaranteed that the sum of n over all test cases does not exceed 1000, and the sum of m over all test cases does not exceed 1000.

All input coordinates and scores are integers. The value of W fits in a signed 64-bit integer.

Output

For each test case, output one real number: the minimum possible value of λ .

Your answer will be considered correct if its absolute or relative error does not exceed 10^{-6} .

More precisely, if your output is x and the correct answer is y , then your answer is accepted if $|x - y| \leq 10^{-6} \cdot \max(1, |y|)$.

Example

standard input	standard output
3	0.0000000000
4 1 5	1.0000000000
-1 -1	2.0000000000
1 -1	
1 1	
-1 1	
0 0 5	
4 2 2	
-1 -1	
1 -1	
1 1	
-1 1	
1 0 1	
0 1 1	
4 2 2	
-1 -1	
1 -1	
1 1	
-1 1	
-2 0 1	
2 0 1	